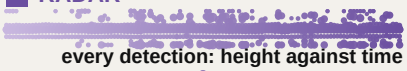
FIVE SENSORS.
ONE DECISION.

Can a machine recognise everyday activities without an ordinary camera?

Multimodal activity recognition under a 100 MB constraint

RADAR

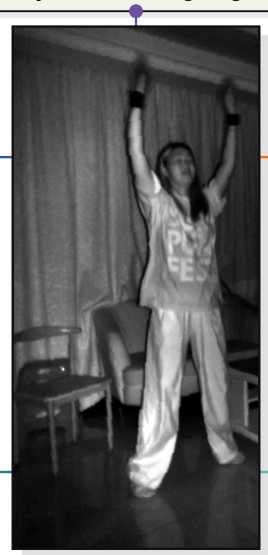


every detection: height against time

IR + DEPTH



body and scene, 640 x 480



THERMAL



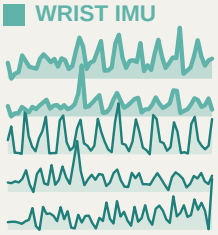
heat of body and objects, 320 x 240

SKELETON



17 joints in 3-D, same instant

WRIST IMU



|acceleration| over the clip

ONE HUMAN ACTION

jumping jacks, as the infrared camera sees it

FIVE SPECIALISTS

body · hands · heat · posture · motion

FUSION

five votes, one prior-corrected log-linear sum

CONTEXT

the order of the recording, from radar timestamps

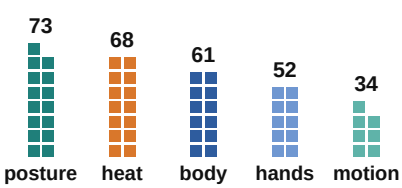
ONE DECISION

jumping jacks: on this held-out clip all five specialists and the fusion say so

40	96 MB	0.877	UNSEEN
ACTIVITIES	ONE FILE	PRIVATE SCORE	PEOPLE AT TEST TIME

14th of 307 teams · 0 / 405 predictions changed across 5 tested GPU generations (posteriors drift up to 2e-3) · no network at inference

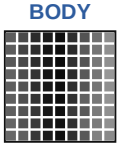
01 DIFFERENT SENSORS RESCUE DIFFERENT MISTAKES



one square = 5 clips only this specialist gets right; 2,399 of 2,700 are right for at least one

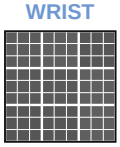
02 MORE PIXELS ARE NOT ALWAYS MORE INFORMATION

BODY



one source pixel = one input pixel

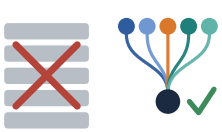
WRIST



one source pixel = nine input pixels

body crop 0.714 to 0.733, 224 to 384 px
wrist crop -1.36 points, 224 to 288 px

03 MORE MODEL IS NOT BETTER GENERALISATION



big model vs five small

- ✗ more capacity
- ✗ training tricks
- ✗ test-time tricks
- ✓ new sensor view
- ✓ better fusion
- ✓ recording order

VideoMAE-B -1.8, Swin3D-T -9.8 points; every lever measured against one bar (paired folds, 2 SE)

Team AGX · Atharv, BITS Pilani, K K Birla Goa Campus · CUHK-X Challenge 2026, Small Model Track

technical report follows